Chapter 1: Introduction

Computer Networking: A Top-Down Approach (Kurose & Ross, 9th edition)

Easy-wording notes covering all 79 slides. The notes follow the slides in order, and slide numbers are given so you can match them with the deck.

Slides 1–2: Title and Chapter Goal

Slide 1 is the title slide and the authors' copyright note. The book is “Computer Networking: A Top-Down Approach” (9th edition) by Jim Kurose and Keith Ross. Nothing to memorize here.

Slide 2 states the goal of the chapter: get the “big picture” and learn the basic terminology. Deeper detail comes in later chapters. The chapter covers these topics:

1 What is the Internet? What is a protocol?

2 Network edge: hosts, access networks, physical media

3 Network core: packet switching and circuit switching, Internet structure

4 Performance: loss, delay, throughput

5 Protocol layers and service models

6 Security

7 History

The “roadmap” slides (9, 22, 44, 55, 62, 73) repeat this list to show which topic is currently being covered.

Slides 3–5: The Internet, the “Nuts and Bolts” View

The “nuts and bolts” view describes the Internet by looking at what it is made of.

Hosts (end systems). These are the billions of devices connected to the Internet: phones, laptops, desktops, servers, cars, fridges, and so on. They run network applications at the Internet's “edge”. The words “host” and “end system” mean the same thing.

Communication links. These connect devices together. They can be fiber, copper, radio, or satellite. How fast a link can send data is called its transmission rate or bandwidth.

Packet switches. These are devices that receive chunks of data, called packets, and forward them onward. The two main types are routers and switches.

Networks. A network is a collection of devices, routers, and links managed by one organization. The slide shows several kinds: home network, enterprise (company) network, mobile network, datacenter network, content provider network, and local/regional and national/global ISPs.

Slide 4: “Fun” devices. This slide shows unusual things that are now online: internet fridges, Amazon Echo, Fitbit, pacemaker monitors, web-enabled toasters, security cameras, cars, bikes, scooters, gaming devices, and AR devices. The point is that the Internet is no longer just computers. This idea is often called the Internet of Things.

Slide 5: Network of networks and protocols.

- The Internet is a “network of networks”. Many separate ISPs (Internet Service Providers) are connected to each other.
- Protocols are everywhere. Examples are HTTP (web), TCP, IP, WiFi, 4G/5G, Ethernet, Zoom, and Skype. Protocols control how messages are sent and received.
- Internet standards are written as RFCs (Request for Comments). They are produced by the IETF (Internet Engineering Task Force). The standards exist so that devices from different companies can work together.

Slide 6: The Internet, the “Services” View

The second way to look at the Internet is by what it does for applications. The Internet is infrastructure that provides services to applications such as the Web, streaming video, video calls, email, games, e-commerce, and social media.

This infrastructure gives programmers a programming interface. You can think of it as “hooks” that let a sending app and a receiving app connect and use the Internet's transport service. A programmer does not have to build the cables or routers. They just use the interface.

The slide compares this to a postal service. The post office offers different service options (normal, express, registered), and you pick one without caring how the mail travels. In the same way, the Internet offers service options to apps.

Slides 7–8: What Is a Protocol?

Human protocols. When you meet someone you say “Hi”, they say “Hi”, and then you ask “What's the time?”. There is a fixed order and a fixed meaning. Introductions and “I have a question” are also protocols.

Network protocols. These are the same idea, but between computers. All communication on the Internet is governed by protocols.

Definition. A protocol defines:

- the format of messages sent and received,
- the order of those messages, and
- the actions taken when a message is sent or received.

In short, it is a set of rules for which messages to send and what to do when a message (or some other event) arrives.

Slide 8 diagram. The human example is: “Hi”, “Hi”, “Got the time?”, “2:00”. The computer example is: TCP connection request, TCP connection response, then GET request for a web page, then the file comes back. Time flows downward in the diagram. Both examples have the same structure: a greeting to set up contact, then a request, then a reply. The slide asks you to think of other human protocols, such as answering a phone call.

Slides 10–12: A Closer Look at Internet Structure

The Internet is split into three parts.

Network edge. This is where hosts live. Hosts are clients and servers. Servers are often kept in data centers.

Access networks and physical media. These are the wired or wireless links that connect edge devices to the first router.

Network core. This is a mesh of interconnected routers. It is the “network of networks” in the middle that carries data across the world.

Slides 10, 11, and 12 build up the same picture one part at a time.

Slide 13: Access Networks and Physical Media

The question is: how do we connect end systems to the edge router? The three types are:

- residential access networks (homes),
- institutional access networks (schools, companies),
- mobile access networks (WiFi, 4G/5G).

Slides 14–15: Cable-Based Access

Cable internet uses the same cable network that carries cable TV.

FDM (frequency division multiplexing). The cable carries many channels at once, and each channel uses a different frequency band. Some channels carry TV video, and some carry data and control. In the diagram, channels 1–9 are video and the last ones are data and control.

Parts. The home has a cable modem and a splitter. The splitter separates TV signals from data signals. The cable goes to the cable headend, where a CMTS (cable modem termination system) connects the cable network to the ISP.

HFC (hybrid fiber coax). The network uses fiber for the long part and coaxial cable for the last part to homes.

Speed. It is asymmetric. Downstream (download) is faster, up to 40 Mbps – 1.2 Gbps. Upstream (upload) is slower, 30–100 Mbps.

Shared. Homes share the access network up to the headend. If your neighbors use a lot, you may get less speed.

Slide 16: DSL (Digital Subscriber Line)

DSL uses your existing telephone line.

- The DSL modem at home sends voice and data at different frequencies over the phone line to the central office.
- At the central office, a DSLAM (DSL access multiplexer) separates them. Voice goes to the telephone network. Data goes to the Internet.
- The line is dedicated to you, not shared like cable.
- Speed: 24–52 Mbps downstream and 3.5–16 Mbps upstream. It is also asymmetric.

Cable vs DSL in one line: cable shares a line with neighbors, while DSL gives each home its own dedicated line to the central office.

Slide 17: Home Networks

A typical home has one box that contains several things:

- a cable or DSL modem,
- a router, a firewall, and NAT,
- a WiFi wireless access point (54 or 450 Mbps).

Devices connect wirelessly through WiFi or by wired Ethernet cable (about 1 Gbps). The box then connects to the cable headend or the central office.

Slide 18: Wireless Access Networks

A shared wireless network connects an end system to a router through a base station, also called an access point. There are two types:

Wide-area cellular networks (4G/5G). These are run by mobile operators. The range is tens of kilometers and the speed is tens of Mbps.

Wireless LANs (WiFi). These are used inside or around a building (about 100 feet). The standards are 802.11b/g/n, with speeds of 11, 54, and 450 Mbps.

Slides 19–21: Links and Physical Media

A **bit** travels from a transmitter to a receiver. The **physical link** is whatever lies between them.

There are two kinds of media:

- **Guided media:** the signal travels inside a solid material (copper, fiber, coax).
- **Unguided media:** the signal travels freely through the air (radio).

Twisted pair (TP). This is two insulated copper wires twisted together. Category 5 supports 100 Mbps and 1 Gbps Ethernet. Category 6 supports 10 Gbps Ethernet.

Fiber optic cable. This is glass fiber that carries light pulses, and each pulse is a bit. It has high speed (tens to hundreds of Gbps) and a low error rate. Repeaters can be placed far apart. It is immune to electromagnetic noise.

Coaxial cable. This has two concentric copper conductors (one inside the other). It is bidirectional and broadband, meaning it carries many frequency channels at once, with hundreds of Mbps per channel.

Wireless radio. The signal travels in bands of the electromagnetic spectrum with no physical wire. It is broadcast and half-duplex (one direction at a time). The environment affects it through reflection, obstruction by objects, and interference or noise.

Radio link types:

- Wireless LAN (WiFi): 10s–100s of Mbps over tens of meters.
- Wide-area (4G/5G): 100s of Mbps over about 10 km.
- Bluetooth: replaces cables over short distances at limited rates.

- Terrestrial microwave: point-to-point, 45 Mbps channels.
- Satellite: under 100 Mbps downlink (Starlink). A geostationary satellite has about 270 ms end-to-end delay because the signal travels very far.

Slide 23: The Network Core

The core is a mesh of interconnected routers. It works by **packet switching**:

- The host breaks application messages into packets.
- The network forwards packets from one router to the next, along links, from source to destination.

Slides 24–26: Two Key Core Functions

Routing is the global action. It decides which path packets should take from source to destination. Routing algorithms do this. A good analogy is planning a trip from start to end before you leave.

Forwarding (also called switching) is the local action. When a packet arrives at a router, the router looks at the destination address in the packet's header and moves the packet to the correct output link. It uses a local forwarding table. The table maps header values to output links. For example, in the slide, a header value of 0111 maps to output link 2.

The analogy for forwarding is passing through one interchange on the trip. Slides 25 and 26 show this with a map: routing is the whole plan, and forwarding is the step at each junction.

Slide 27: Host Sends Packets

A host does the following:

1. Takes an application message.
2. Breaks it into smaller chunks called packets, each of length L bits.
3. Sends each packet into the access network at the transmission rate R.

R is also called link capacity or link bandwidth.

Packet transmission delay = L / R. This is the time needed to push all L bits of a packet onto the link.

Slide 28: Store-and-Forward

Store-and-forward means a router must receive the entire packet before it can start sending it on the next link.

The transmission delay for one hop is L/R seconds.

Example. L = 10 Kbits and R = 100 Mbps. So 10,000 / 100,000,000 = 0.0001 seconds = 0.1 msec per hop.

Slides 29–30: Queueing and Packet Loss

Each router has an output link with a queue (buffer) of packets waiting to be sent.

- Queueing happens when packets arrive faster than the link can send them. In the diagram, fast links (100 Mb/s) feed into a slow link (1.5 Mb/s), so packets pile up.
- If the arrival rate is higher than the transmission rate for some time, packets queue up.
- The buffer has limited memory. If it fills up, new packets are dropped (lost).

Slide 31: Circuit Switching

Circuit switching is the alternative to packet switching. It is used in old telephone networks.

- Before the call, resources are reserved along the whole path from source to destination.
- The resources are dedicated. There is no sharing.
- Performance is guaranteed, like a real circuit.
- The downside is that if the user is silent, the reserved circuit sits idle and cannot be used by anyone else.

In the diagram, each link has four circuits. A call gets the 2nd circuit on the top link and the 1st circuit on the right link.

Slide 32: FDM and TDM

These are two ways to split a link into circuits.

FDM (Frequency Division Multiplexing). The frequency range is divided into narrow bands. Each call gets its own band and can transmit at the maximum rate of that narrow band, all the time.

TDM (Time Division Multiplexing). Time is divided into slots. Each call gets periodic slots and can transmit at the maximum rate of the full (wider) band, but only during its own slots.

Simple way to remember: FDM splits by frequency (different lanes), TDM splits by time (taking turns).

Slides 33–34: Packet Switching vs Circuit Switching

Example. A 1 Gb/s link is shared. Each user sends at 100 Mb/s when active, but is active only 10% of the time.

- Circuit switching: 1000 / 100 = 10 users only, because each user reserves 100 Mb/s even when idle.
- Packet switching: with 35 users, the chance that more than 10 are active at the same time is less than 0.0004. So many more users can share the same link.

Is packet switching a clear winner? Mostly, but not always.

Advantages:

- Great for “bursty” data (sometimes sending, sometimes quiet).
- Resource sharing.
- Simpler, with no call setup.

Disadvantages:

- Excessive congestion is possible, causing packet delay and loss from buffer overflow.

- Protocols are needed for reliable data transfer and congestion control.

The slide asks how to get circuit-like behavior from packet switching. The answer is “it's complicated”, and later chapters study techniques for this. The human analogy for the two types: a restaurant where you reserve a table (circuit switching) versus walking in and waiting for a free table (packet switching).

Slides 35–43: Internet Structure, a Network of Networks

This is a step-by-step story of how the Internet's structure developed.

Slide 35 (setup). Hosts connect through access ISPs. These access ISPs must be connected to each other so any two hosts can exchange packets. The result is very complex. It evolved because of economics and national policies.

Slides 36–37 (Step 1: connect everyone directly). Millions of access ISPs exist. Connecting each pair directly does not scale. It needs about N^2 connections, which is too many.

Slide 38 (Step 2: one global ISP). Connect each access ISP to one global transit ISP. The customer ISP and the provider ISP have an economic agreement (customer pays provider).

Slide 39 (Step 3: competitors). If one global ISP is profitable, competitors (ISP A, B, C) will appear.

Slide 40 (Step 4: IXPs). Competing ISPs need to be connected to each other. An Internet exchange point (IXP) is a meeting place where ISPs connect. A direct connection between two ISPs is called a peering link.

Slide 41 (Step 5: regional ISPs). Regional networks can arise to connect access networks to the larger ISPs.

Slide 42 (Step 6: content providers). Companies like Google, Microsoft, and Akamai run their own private networks to bring content close to users.

Slide 43 (final picture). At the center is a small number of large, well-connected networks:

- Tier-1 commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT) with national and international coverage.
- Content provider networks (e.g., Google, Facebook) that connect their data centers to the Internet, often bypassing tier-1 and regional ISPs.

Slides 45 and 51: How Delay and Loss Happen

Packets wait in router buffers for their turn. The queue grows when the arrival rate is temporarily higher than the output link capacity. If the buffer is full, arriving packets are dropped. This is packet loss.

On slide 51: a lost packet may be resent by the previous node, by the source host, or not at all.

Slides 46–47: Four Sources of Packet Delay

Total delay at one node:

$$\mathbf{d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}}$$

1. **d_{proc} (nodal processing).** The router checks for bit errors and decides the output link. It takes very little time, usually under microseconds.
2. **d_{queue} (queueing delay).** Time spent waiting in the queue for transmission. It depends on how congested the router is.
3. **d_{trans} (transmission delay).** Time to push all bits of the packet onto the link. $d_{trans} = L / R$, where L is packet length in bits and R is link rate in bps.
4. **d_{prop} (propagation delay).** Time for a bit to travel along the link. $d_{prop} = d / s$, where d is link length and s is propagation speed (about 2×10^8 m/s).

Important: d_{trans} and d_{prop} are very different. Transmission delay depends on packet size and link rate. Propagation delay depends on distance.

Slides 48–49: Caravan Analogy

Think of a caravan of 10 cars on a highway with toll booths.

- Car = bit
- Caravan = packet
- Toll booth = link (its service time = transmission)
- Driving between toll booths = propagation

Slide 48. The toll booth takes 12 seconds per car, and cars travel at 100 km/hr. The toll booths are 100 km apart.

- Time to push the whole caravan through the toll booth = $12 \times 10 = 120$ seconds.
- Time for the last car to travel 100 km = 1 hour.
- Total = 62 minutes.

Slide 49. Now cars travel at 1000 km/hr and the toll booth takes 1 minute per car. After 7 minutes, the first car reaches the second booth while 3 cars are still waiting at the first booth. So yes, cars can arrive at the second booth before all cars are serviced at the first. This shows that the first bits of a packet can reach the next node before the whole packet has been sent.

Slide 50: Queueing Delay Revisited

Traffic intensity = $L \cdot a / R$

- a = average packet arrival rate
- L = packet length (bits)
- R = link bandwidth (bits/sec)

It compares the rate bits arrive to the rate bits can be served.

- Near 0: average queueing delay is small.
- Near 1: average queueing delay becomes large.
- Greater than 1: more work arrives than can be served, so delay grows without limit (infinite).

Slides 52–54: Throughput

Throughput is the rate (bits per unit time) at which bits are successfully delivered from sender to receiver.

- Instantaneous: the rate at one moment.
- Average: the rate over a longer time.

The slide uses a pipe analogy: the server sends fluid (bits) into pipes with different capacities. A server sends a file of F bits to a client. Rs is the server link rate and Rc is the client link rate.

- If Rs < Rc, throughput is Rs.
- If Rs > Rc, throughput is Rc.

In other words, throughput is limited by the slowest link. That slowest link is the **bottleneck link**.

Slide 54 (network scenario). 10 connections share a backbone link of rate R. Each connection's throughput is:

min(Rc, Rs, R/10)

In practice, Rc or Rs (the access links) is usually the bottleneck.

Slides 56–61: Network Security

Slides 56–57. The Internet was not designed with much security in mind. The original idea was a group of mutually trusting users on a transparent network. Now designers are playing catch-up. We must think about how attackers attack networks, how to defend, and how to design architectures that are immune to attacks.

Slide 58: Packet sniffing. On broadcast media (shared Ethernet, wireless), a device can put its network card in promiscuous mode and record all packets passing by, including passwords. Wireshark is a free packet-sniffer tool used in the book's labs.

Slide 59: IP spoofing. An attacker sends a packet with a false source address, pretending to be someone else.

Slide 60: Denial of Service (DoS). Attackers overwhelm a server or bandwidth with fake traffic, so real users cannot get service. The usual steps are:

1. Select a target.
2. Break into many hosts across the network (these form a botnet).
3. Send packets to the target from all the compromised hosts at once.

When many machines do it together, it is called a distributed attack (DDoS).

Slide 61: Lines of defense.

- **Authentication:** proving you are who you say you are. Cellular networks use the SIM card as hardware identity. The traditional Internet has no such hardware help.
- **Confidentiality:** done through encryption.
- **Integrity checks:** digital signatures prevent or detect tampering.
- **Access restrictions:** password-protected VPNs.
- **Firewalls:** specialized “middleboxes” that are off by default. They filter incoming packets to restrict senders, receivers, and applications, and they help detect and react to DoS attacks.

Security is covered in depth in Chapter 8.

Slides 63–65: Protocol Layers

Networks are complex, with many pieces: hosts, routers, links, applications, protocols, hardware, and software. Layering is a way to organize this.

Why layering (slide 64)?

- Explicit structure helps us identify the pieces and how they relate.
- Modularization makes maintenance and updating easier.
- Changing how one layer works is invisible to the rest of the system. The slide's example: changing the gate procedure at an airport does not affect the rest of the airline system.

The Internet protocol stack (5 layers, top to bottom):

1. **Application:** supports network applications. Examples: HTTP, IMAP, SMTP, DNS.
2. **Transport:** process-to-process data transfer. Examples: TCP, UDP.
3. **Network:** routing of datagrams from source to destination. Examples: IP and routing protocols.
4. **Link:** data transfer between neighboring network elements. Examples: Ethernet, 802.11 (WiFi), PPP.
5. **Physical:** bits “on the wire”.

A memory trick for the order is “All People Seem To Need Data Processing” (Application, Presentation, Session, Transport, Network, Data link, Physical) for the 7-layer model. For the 5-layer Internet stack just drop Presentation and Session.

Slides 66–71: Services, Layering, and Encapsulation

Each layer adds its own header to the data coming from the layer above. This wrapping is called **encapsulation**.

Going down from the source:

1. The application creates a **message (M)**.
2. The transport layer adds header H_t, creating a **segment** [H_t | M]. H_t is used by the transport protocol to provide its service (for example, reliable delivery).
3. The network layer adds header H_n, creating a **datagram** [H_n | H_t | M].
4. The link layer adds header H_l, creating a **frame** [H_l | H_n | H_t | M].
5. The physical layer sends the bits.

At the destination, the process runs in reverse: each layer removes its own header and passes the rest up, until the application gets the original message M.

Slide 69 analogy. Matryoshka (Russian stacking) dolls. A small doll sits inside a bigger one, just as the message sits inside the segment, which sits inside the datagram, which sits inside the frame.

Slide 71: end-to-end view. The source host uses all five layers. A router in the middle uses only the bottom three (network, link, physical), because it only needs to read the network header to forward. A link-layer switch uses only the bottom two (link, physical). The destination host uses all five layers

again.

Slide 72: ISO/OSI Reference Model

The OSI model has seven layers. It has two extra layers that the Internet stack does not have:

- **Presentation:** lets applications interpret the meaning of data (encryption, compression, machine-specific formats).
- **Session:** synchronization, checkpointing, and recovery of data exchange.

In the Internet stack, these services, if needed, must be implemented inside the application. The slide asks whether they are really needed.

The seven layers are: application, presentation, session, transport, network, link, physical.

Slides 74–78: Internet History

1961–1972: early packet-switching.

- 1961: Kleinrock showed with queueing theory that packet switching works well.
- 1964: Baran proposed packet switching for military networks.
- 1967: ARPAnet was conceived by ARPA (Advanced Research Projects Agency).
- 1969: first ARPAnet node working.
- 1972: public demo of ARPAnet, with NCP (Network Control Protocol) as the first host-to-host protocol and the first email program. ARPAnet had 15 nodes.

1972–1980: internetworking and new networks.

- 1970: ALOHAnet, a satellite network in Hawaii.
- 1974: Cerf and Kahn designed the architecture for interconnecting networks.
- 1976: Ethernet at Xerox PARC.
- Late 1970s: proprietary architectures such as DECnet, SNA, and XNA.
- 1979: ARPAnet had 200 nodes.
- Cerf and Kahn's principles, which define today's Internet architecture: minimalism and autonomy (no internal changes needed to connect networks), best-effort service model, stateless routing, and decentralized control.

1980–1990: new protocols and more networks.

- New national networks: CSnet, BITnet, NSFnet, Minitel.
- 100,000 hosts connected.
- 1982: SMTP email protocol defined.
- 1983: TCP/IP deployed. DNS defined (name to IP address translation).
- 1985: FTP defined.
- 1988: TCP congestion control.

1990s–2000s: commercialization and the Web.

- Early 1990s: ARPAnet shut down.

- 1991: NSF lifted the restrictions on commercial use of NSFnet (shut down in 1995).
- Early 1990s: the Web. It was built on the hypertext idea (Bush 1945, Nelson 1960s). HTML and HTTP were created by Berners-Lee.
- 1994: Mosaic, later Netscape.
- Late 1990s: Web commercialized. More “killer apps” such as instant messaging and P2P file sharing. Network security came to the front.
- About 50 million hosts, 100 million+ users, backbone links at Gbps speeds.

2005–present: scale, SDN, mobility, cloud.

- Aggressive rollout of broadband home access (10–100s of Mbps).
- 2008: software-defined networking (SDN).
- High-speed wireless access everywhere (4G/5G, WiFi).
- Google, Facebook, Microsoft built their own networks to reach users directly.
- Enterprises run services in the cloud (Amazon Web Services, Microsoft Azure).
- Smartphones: by 2017 more mobile than fixed devices were on the Internet.
- About 15 billion devices attached in 2023.

Slide 79: Chapter Summary

Chapter 1 covered: Internet overview, what a protocol is, network edge, access network and core, packet vs circuit switching, Internet structure, performance (loss, delay, throughput), layering and service models, security, and history. You now have the vocabulary and the “feel” of networking. More depth follows in later chapters.

Quick Formula Sheet

- Packet transmission delay: $d_{trans} = L / R$
- Propagation delay: $d_{prop} = d / s$
- Total nodal delay: $d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$
- Traffic intensity: $L \cdot a / R$
- End-to-end throughput with shared bottleneck: min(Rc, Rs, R/10)

Terms to Remember

Host, end system, packet, router, switch, ISP, protocol, RFC, IETF, access network, edge, core, bandwidth, FDM, TDM, HFC, CMTS, DSL, DSLAM, store-and-forward, queueing delay, packet loss, bottleneck link, IXP, peering, tier-1 ISP, sniffing, spoofing, DoS, botnet, firewall, encapsulation, segment, datagram, frame.

FIN